

AWS Cloud Engineering Lecture Notes

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1. Amazon DynamoDB:

- DynamoDB is a fully managed NoSQL database service that provides fast and predictable performance with seamless scaling.
- It supports key-value and document data structures.
- DynamoDB partitions data across physical servers to meet scalability needs. It uses the partition key value as input to an hash function to determine the partition.
- Consistent Reads: Strongly consistent reads return a result that reflects all writes that received a successful response prior to the read.
- DynamoDB Streams captures a time-ordered sequence of item-level modifications in any DynamoDB table.

2. Amazon S3 (Simple Storage Service):

- S3 is an object storage service offering industry-leading scalability, data availability, security, and performance.
- S3 buckets are resources where objects are stored. Buckets have a globally unique name.
- S3 Storage Classes: Standard, Intelligent-Tiering, Standard-IA, One Zone-IA, Glacier Instant Retrieval, Glacier Flexible Retrieval, and Glacier Deep Archive.
- Lifecycle policies can be used to automatically transition objects to cheaper storage classes or expire them.

3. AWS Lambda:

- AWS Lambda is a serverless, event-driven compute service that lets you run code for virtually any type of application or backend.
- Lambda automatically scales your application by running code in response to each event.
- Maximum execution timeout for a Lambda function is 15 minutes (900 seconds).